

Gen Li

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[GitHub](#) | [Google Scholar](#) | [Homepage](#)

RESEARCH INTERESTS

My research interests lie in **multimodal reasoning and agentic systems**, with a focus on enabling large multimodal models to understand complex environments, leverage external knowledge and tools, and perform reliable and computationally efficient reasoning through post-training and inference-time optimization.

EDUCATION

- **University of Electronic Science and Technology of China** Sep. 2021 – Jun. 2024
M.Eng. in Information and Communication Engineering · GPA: 3.91/4.00 Chengdu, China
- **Sichuan University** Sep. 2017 – Jun. 2021
B.Eng. in Communication Engineering · GPA: 3.72/4.00 Chengdu, China
 - Selected coursework: Calculus (95/100), Probability and Statistics (94/100), Complex Analysis and Integral Transforms (93/100).
 - Selected for recommendation-based graduate admission with exemption from the National Postgraduate Entrance Examination (top ~10% of the undergraduate cohort).

PUBLICATIONS

- [1] **Gen Li**, Peiyu Liu. *FastV-RAG: Towards Fast and Fine-Grained Video QA with Retrieval-Augmented Generation*. In **Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)**. [Paper] [Code]
- [2] **Gen Li**, X. Wang, C. Zhao, et al. *Ring Sketch: A Generic, Low-Complexity, and Hardware-Friendly Traffic Measurement Framework over Sliding Windows*. In **Proceedings of the IEEE International Conference on Communications (ICC 2024)**. [Paper] [Code]

RESEARCH EXPERIENCE

- **FastV-RAG: Efficient Multimodal RAG for Knowledge-Intensive Video QA** Mar. 2024 – Oct. 2025
Vision-Language Models, Retrieval-Augmented Generation, Speculative Decoding ACL 2026
 - Developed *VideoSpeculateRAG*, a parallel draft–verify framework in which a 3B VLM generates per-document candidate answers in parallel and a 32B VLM jointly scores their reliability in a single forward pass. The framework shifts retrieval-conditioned generation to the faster lightweight model, while the large verifier evaluates only the short candidate outputs rather than generating from the full retrieved context, reducing both context length and generation latency.
 - Diagnosed two recurring retrieval-induced entity errors: *Cross-Entity Transfer*, where the model grounds the correct visual entity but applies knowledge from a similar retrieved entity, and *Entity Substitution*, where it fully follows an incorrect entity. Introduced explicit entity extraction and frame-level vision–language alignment to verify whether each candidate answer remained grounded in the video before final selection.
 - Conducted controlled ablations on verifier reliability, entity alignment, selection-threshold sensitivity, retrieval quality, and alternative encoders. The results showed that reliability scoring and visual alignment captured complementary failure modes and were both necessary for robust candidate selection.
 - Maintained comparable accuracy to standard 32B RAG on VideoSimpleQA (**91.12% vs. 91.30%**) and surpassed it on Encyclopedic VQA (**47.64% vs. 46.50%**), while reducing end-to-end latency by **46.06%** and **58.90%**, respectively; validated the method with Qwen2.5-VL and Qwen3-VL configurations.
- **Reinforcement Learning for Collaborative Network Measurement** Sep. 2023 – Mar. 2024
Deep Reinforcement Learning, Graph Neural Networks, Network Measurement Master’s Thesis
 - Formulated collaborative flow measurement in resource-constrained data center networks as a sequential decision-making problem.
 - Developed a topology-aware deep Q-learning framework using a graph attention network and a composite reward based on predicted measurement error.
 - Reduced maximum mean relative error by approximately **50%** compared with hash-based coordination baselines on a 45-node fat-tree topology.

WORK EXPERIENCE

- **Baidu Inc.** Jul. 2024 – Present
Research Engineer, Full-Duplex Multimodal Voice Agents Beijing, China
 - **Training a cascaded ASR–LLM–TTS voice agent.** Built the dialogue-model training pipeline for a production-scale AI phone agent serving local-service merchants. Designed a dual-agent simulation framework in which a caller agent modeled conversational styles, user intents, and profile information from anonymized real-world calls, while a DeepSeek-R1-based service agent generated reasoning trajectories and responses under merchant-specific business constraints. Constructed more than **500K** synthetic multi-turn dialogues and performed offline reasoning distillation by fine-tuning a Qwen3-8B backbone on DeepSeek-R1-generated reasoning trajectories and responses, enabling explicit reasoning over user intent, dialogue context, and business constraints.
 - **Efficient reasoning via length-aware reinforcement learning.** Reduced cascaded ASR–LLM–TTS latency from over **3.3 s** to below **2.0 s** through streaming execution. Identified autoregressive reasoning generation as the primary bottleneck and designed a piecewise GRPO reward that enforced a preferred reasoning-length range and down-weighted overlong traces, reducing reasoning length by **40%** and model latency by **500 ms** without regression on internal dialogue-quality evaluations.
 - **Audio-native multimodal modeling with progressive post-training.** To address cascade latency, cross-module error accumulation, and information loss from intermediate text representations, co-developed a Qwen3-Omni-based audio-native dialogue system that jointly handled speech understanding, dialogue reasoning, turn-taking prediction, and response generation. Designed a staged SFT and RL curriculum that first established domain and interaction capabilities, and then adapted the model to complex real-world calls using large-scale dialogue data and chain-of-thought supervision, improving the human-rated high-quality dialogue rate by **29.3 percentage points**.
 - **Dynamic reasoning control via reinforcement learning.** Found that always-on chain-of-thought increased time-to-first-token from **0.4 s** to **1.6 s**, although over half of dialogue turns involved routine scenarios that did not require deep reasoning. Developed an RL-based policy that dynamically selected between direct response and explicit reasoning for each turn, reducing end-to-end latency by an additional **0.3 s**.

HONORS AND AWARDS

- **Outstanding Graduate Student**, University of Electronic Science and Technology of China 2024
- **Second-Class Academic Scholarship**, University of Electronic Science and Technology of China 2022 – 2023
- **Certificate of Excellence in Engineering**, Sichuan University 2021
- **First-Class Academic Scholarship**, Sichuan University 2018 – 2019
- **Second-Class Academic Scholarship**, Sichuan University 2017 – 2018

TEACHING EXPERIENCE

- **University of Electronic Science and Technology of China** Sep. 2022 – Jun. 2023
Teaching Assistant, Data Structures and Algorithms Chengdu, China
 - Led weekly office hours and laboratory sessions, graded programming assignments, and provided technical support to undergraduate students.

ACADEMIC SERVICE

- **Reviewer**, NeurIPS 2026 2026

TECHNICAL SKILLS

- **Programming:** Python, C++, Java, SQL, Hive/SparkSQL, Bash
- **Frameworks and Systems:** PyTorch, Hugging Face Transformers, vLLM, ms-swift
- **Training and Post-Training:** Supervised fine-tuning, GRPO, DPO, LoRA, reward design
- **Multimodal Modeling:** Vision-language models, multimodal agents, retrieval-augmented generation, speculative decoding